



CP Bootcamp 2026 - Day 16

Theme: Final Contest Preparation + CP Strategy + Full Revision

Main Goal:

আজকে নতুন algorithm শেখার দিন না।

আজকে তুমি গত ১৫ দিনের সব knowledge একসাথে combine করবে এবং **Contest**

Programmer mindset তৈরি করবে।

আজকের focus:

Knowledge

↓

Pattern Recognition

↓

Problem Selection

↓

Implementation

↓

Debugging

↓

Contest Strategy

Estimated Time: 6–7 hours



Day 16 Chapter Map

Day 16

- Chapter 1 - CP Journey Review
- Chapter 2 - Problem Reading Strategy
- Chapter 3 - Pattern Recognition System
- Chapter 4 - Implementation Strategy
- Chapter 5 - Debugging & Testing System
- Chapter 6 - Contest Time Management
- Chapter 7 - CP Template & Environment Setup
- Chapter 8 - Virtual Contest Practice
- Chapter 9 - After Bootcamp Roadmap
- Chapter 10 - Final Assignment & Graduation

Chapter 1 - CP Journey Review

গত ১৬ দিনে তুমি যে journey complete করেছো:

Day 0

Environment + Mindset

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Day 1

C + Basic Problem Solving

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Day 2

Array Patterns

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Day 3

String Patterns

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Day 4

Nested Loop + Pattern Thinking

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Day 5

Sorting + Frequency

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Day 6

Functions + Modular Code

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Day 7

Searching

↓

Day 8

Sorting Algorithms + Complexity

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Day 9

Two Pointer + Sliding Window

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Day 10

Recursion + Backtracking

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Day 11

Stack + Queue

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Day 12

Linked List + Hashing

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Day 13

Trees

↓

Day 14

Graphs

↓

Day 15

Dynamic Programming

↓

Day 16

Contest Mindset

এখন তোমার কাছে কী আছে?

তুমি এখন problem দেখলে চিন্তা করবে:

Problem

↓

Which pattern?

↓

Which data structure?

↓

Which algorithm?

↓

Implementation

Chapter 2 - Problem Reading Strategy

Contest-এ সবচেয়ে বড় skill:

Code লেখা না, problem বুঝা।

Step 1 - Read Problem Carefully

প্রথমে দেখো:

Input

কি দেওয়া আছে?

Example:

n

array

string

graph

Output

কি চাচ্ছে?

Example:

maximum

minimum

count

index

yes/no

Constraints

সবচেয়ে important।

Example:

$$1 \leq n \leq 100000$$

মানে:

$O(n^2)$ চলবে না।

Need:

$$O(n)$$

$$O(n \log n)$$

Step 2 - Identify Keywords

কিছু keyword দেখলে pattern চিনবে।

"Count"

Think:

Frequency Array

Hashing

"Maximum / Minimum"

Think:

Greedy

DP

Sorting

"Sorted"

Think:

Binary Search

Two Pointer

"Subarray / Substring"

Think:

Sliding Window

Prefix Sum

"All Possible"

Think:

Recursion

Backtracking

"Connected"

Think:

Graph

DFS

BFS

Chapter 3 - Pattern Recognition System

Contest-এ প্রথমে algorithm না।

Pattern চিনতে হবে।

Pattern 1 - Traversal

Question:

"সব element দেখো"

Use:

Loop

Pattern 2 - Frequency

Question:

"কতবার এসেছে?"

Use:

Array Frequency

Hash Map

Pattern 3 - Searching

Question:

"খুঁজে বের করো"

Use:

Linear Search

Binary Search

Pattern 4 - Sorting

Question:

"Order দরকার"

Use:

Sorting

Pattern 5 - Two Pointer

Question:

"Pair"

"Sorted array"

Use:

Two Pointer

Pattern 6 - Sliding Window

Question:

"Continuous"

"Subarray"

Use:

Sliding Window

Pattern 7 - Recursion

Question:

"Break into smaller same problem"

Use:

Recursion

Pattern 8 - Graph

Question:

"Connection"

Use:

DFS/BFS

Pattern 9 - DP

Question:

"Previous answer needed"

Use:

Dynamic Programming

Chapter 4 - Implementation Strategy

Contest-এ code লেখার আগে:

Step 1

Brute Force চিন্তা করো।

Example:

Problem:

Find duplicate.

Brute:

Compare every pair

Complexity:

$O(n^2)$

Step 2

Optimize করো।

Think:

Can I store information?

Answer:

Hashing

Step 3

Template use করো।

Example:

Binary Search:

```
while(left ≤ right)
{
    mid = left+(right-left)/2;

    ...
}
```

Chapter 5 - Debugging & Testing System

Professional programmer:

Sample input-এর উপর শুধু test করে না।

Edge Case চিন্তা

Empty

$n=0$

One Element

$n=1$

Duplicate

5 5 5 5

Sorted

1 2 3 4

Reverse Sorted

4 3 2 1

Large Value

1000000000

নিজের Test বানানোর Habit

Example:

Problem:

Maximum number.

Sample:

5
1 4 7 2 3

নিজে:

1

10

All same:

5

7 7 7 7 7

Negative:

-5 -2 -8

Chapter 6 - Contest Time Management

ধরো contest:

2 hours.

Strategy:

First 10 Minutes

সব problem পড়ো।

Mark:

Easy

Medium

Hard

Solve Order

Always:

Easy

↓

Medium

↓

Hard

Wrong approach:

Hard problem নিয়ে ১ ঘণ্টা বসে থাকা।

20 Minute Rule

যদি ২০ মিনিটেও idea না আসে:

Leave

অন্য problem solve করো।

Debug Time

শেষ ১৫ মিনিট:

Check

Submit

Recheck

Chapter 7 - CP Template & Environment Setup

তোমার repository:

CP-Bootcamp-2026/

└ Templates/

└ HackerRank/

└ CSES/

└ Codeforces/

└ Contest/

└ Notes/

Template রাখবে:

Templates/

└ template.c

Basic Template:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    return 0;
```

```
}
```

Contest-এর আগে check:

```
gcc template.c -o template
```

```
./template
```

Chapter 8 - Virtual Contest Practice

এখন থেকে real practice:

Week Plan

Day 1

Easy Problems

5 problems

Day 2

Mixed:

3 Easy

2 Medium

Day 3

Virtual Contest:

2 hours

No help

Contest Simulation Rules

During contest:

✗ Google না

✗ AI না

✗ Solution দেখা না

After contest:

Analyze:

Why failed?

What pattern missed?

What concept missing?

Chapter 9 - After Bootcamp Roadmap

Bootcamp শেষ।

এখন next phase:

Phase 1 - Problem Practice

Platforms:

HackerRank

CSES

Codeforces

Goal:

300-500 problems

Phase 2 - Algorithms Deep Dive

Learn:

Prefix Sum

Greedy

Advanced Graph

Advanced DP

Number Theory

Bit Manipulation

Phase 3 - Contest

Participate:

Codeforces Div 4

Codeforces Div 3

University Contest

Phase 4 - Competitive Programmer Mindset

Develop:

Observation

Speed

Accuracy

Debugging

Chapter 10 - Final Assignment & Graduation

Part A - Create Your CP Profile

Complete:

☐ Codeforces Account

☐ CSES Account

☐ HackerRank Account

☐ GitHub CP Repository

Part B - Final Project

Create:

CP-Portfolio/

Structure:

CP-Portfolio/

├─ c/

├─ Arrays/

├─ Strings/

├─ Searching/

├─ Sorting/

├─ Graph/

├─ DP/

├─ Templates/

└─ README.md

Part C - Final Contest

Do:

একটি Virtual Contest:

Time:

2 hours

Solve:

Minimum 3 problems

Part D - Self Evaluation

নিজেকে প্রশ্ন:

Problem Solving

আমি কি:

Problem বুঝতে পারি?

Pattern চিনতে পারি?

Complexity ভাবতে পারি?

Coding

আমি কি:

Bug খুঁজে বের করতে পারি?

Clean code লিখতে পারি?

Template ব্যবহার করতে পারি?



Glossary Update

Add:

Brute Force

Optimization

Constraint

Time Complexity

Space Complexity

Pattern Recognition

Virtual Contest

Contest Strategy

Edge Case

Debugging

Implementation



Pattern Library Update

Add:

Problem Solving Framework

Read Problem



Understand Constraints



Find Pattern



Choose Data Structure



Implement



Test Edge Cases



Submit

Contest Strategy

Easy Problems First



Secure Points



Medium Problems



Hard Problems



Review

Day 16 Completion Checklist

- [] Problem reading strategy বুঝি
- [] Constraint দেখে complexity choose করতে পারি
- [] Pattern চিনতে পারি
- [] Edge case test করতে পারি
- [] Contest time manage করতে পারি
- [] Virtual contest দিতে পারি
- [] CP template ready আছে
- [] Next roadmap জানি

CP Bootcamp 2026 - Completed

Final Skill Tree:

C Programming

↓

Problem Solving

↓

Arrays & Strings

↓

Searching & Sorting

↓

Optimization

↓

Recursion

↓

Data Structures

↓

Graphs

↓

Dynamic Programming

↓

Contest Strategy

Bootcamp শেষ।

এখন তোমার কাজ:

Learn Less

Practice More

Analyze Mistakes

Repeat

পরবর্তী phase:

The Engineer's Blueprint Journey → Engineering Mindset + CS Foundation 🚀